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Wishart Distribution

In Univariate Statistics, What dist^n do we get when add squared standard normal obs?

Suppose,

$$Z_1, Z_2 - - , Z_n \sim N(0,1) \text{ indpendly.}$$

Then,

$$Z_1^2 + Z_2^2 + - + Z_n^2 \sim \chi^2_n \quad \text{d.f}$$

$$\boxed{\sum_{i=1}^{n} Z_i^2 \sim \chi^2_n} \rightarrow \text{Write on Board}$$

What if instead of one variable, we have several normally distributed variables?

This is the key Transition.

Move from one var. to several variables –

Suppose, $X_i = \begin{bmatrix} X_{i1} \\ X_{i2} \\ \vdots \\ X_{ip} \end{bmatrix}$ is a p-dim random vector.

Assume, $X_i \sim N_p(0, \Sigma)$

Now, instead of calculating one quantity such as $\sum_{i=1}^{n} Z_i^2$, we need to calculate something involving squares & cross products of several variables.

For eg: , with two variables –

$$\Sigma X_i^2, \quad \Sigma Y_i^2, \quad \Sigma X_i Y_i$$

Introduce the Cross-Product matrix :–

For observations, $X_1, X_2 \dots, X_n$

write,

$$A = \sum_{i=1}^{n} X_i X_i'$$

This is a $p \times p$ matrix.

For $p = 2$, $X_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix}$

Then $X_i X_i' = \begin{pmatrix} x_i^2 & x_i y_i \\ x_i y_i & y_i^2 \end{pmatrix}$

Therefore,

$$A = \begin{bmatrix} \Sigma x_i^2 & \Sigma x_i y_i \\ \Sigma x_i y_i & \Sigma y_i^2 \end{bmatrix}$$

this matrix contains the sum of squares & cross-products. In multivariate analysis the matrix plays a role similar to the sum of squares in univariate.

Now Introduce Wishart Distribution :–

Just as the Chi-square dist^n describes the sum of squared normal variables, the Wishart dist^n describes the dist^n of this matrix of sum of squares & cross products when

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the observations come from a MND.

$X_i \sim N_p(0, \Sigma)$

Write:

$$A = \sum_{i=1}^{n} X_i X_i' \sim W_p(n, \Sigma)$$

Zero mean multivariate normal vector.

p = dim of the random vector

n = degrees of freedom

Σ = covariance matrix

A = random $p \times p$ symmetric matrix.

For $p = 1$ → The Wishart dist^n becomes the chi-square dist^n.

"Wishart dist^n is a multivariate generalization of the chi-square dist^n."

Why do we need ~~this~~ Wishart Dist^n?

Suppose, $X_1, X_2 \dots X_n \sim N_p(M, \Sigma)$

Then Sample Cov^n matrix is —

$$S = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})(X_i - \bar{X})'$$

Because we have used 1 d.f to estimate the unknown mean

$$(n-1)\, S \sim W_p(n-1, \Sigma)$$

$\hookrightarrow A = \sum_{i=1}^{n} (X_i - M)(X_i - M)' \sim W_p(n, \Sigma)$

On Univariate $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$

Properties of Wishart Dist^n:-

① If A_1 is dist^n as $W_{m_1}(A_1 | \Sigma)$ independently of A_2, which is dist^n as $W_{m_2}(A_2 | \Sigma)$

then $A_1 + A_2$ is dist^n as $W_{m_1+m_2}(A_1+A_2 | \Sigma)$

That is d.f add.

Proof:-

$$A = \sum_{i=1}^{n} X_i X_i' \sim W_p(n, \Sigma)$$

where

$X_i \sim N_p(0, \Sigma)$ independtly.

Basic operation on it – Perform some

① Adding two independent Wishart matrices.

② Tranforming a Wishart matrix by a matrix C

③ Taking a quadratic for $a'Aa$.

P-1 Addition of independent Wishart Matrices

If $A_1 \sim W_p(n_1, \Sigma)$

and $A_2 \sim W_p(n_2, \Sigma)$ are independt

then, $A_1 + A_2 \sim W_p(n_1+n_2, \Sigma)$.

Degrees of Freedom add

$X \sim \chi^2_{n_1}$, $Y \sim \chi^2_{n_2}$

$X+Y \sim \chi^2_{n_1+n_2}$ Similar to this

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"Wishart property is the matrix version of the chi-square addition property."

P-2 Transformation of a Wishart Matrix:-

If $A \sim W_p(n, \Sigma)$
& C is a fixed $q \times p$ matrix then,

$$CAC' \sim W_q(n, C\Sigma C')$$

Start From: $A = \sum_i X_i X_i'$

Now Calculate $CAC' = C\left(\sum_{i=1}^{n} X_i X_i'\right) C'$

$$= \sum_i C X_i X_i' C' \quad (\text{Move C inside})$$

Now $C X_i X_i' C' = (CX_i)(CX_i)'$

$$\therefore CAC' = \sum_{i=1}^{n} (CX_i)(CX_i)'$$

Now, Since $X_i \sim N_p(0, \Sigma)$
a linear transformation gives,

$$CX_i \sim N_q(0, C\Sigma C')$$

$$\therefore \sum_i (CX_i)(CX_i)' \sim W_q(n, C\Sigma C')$$

Hence, $\boxed{CAC' \sim W_q(n, C\Sigma C')}$

Explain what C is doing

Suppose, $X = [X_1 \; X_2 \; X_3]'$

If we choose, $C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

then

$$CX = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

So, the original vector has dim $p=3$, but after transformation we have dim $q=2$

Therefore the Wishart changes from W_3 to W_2.

P-3 Quadratic form:-

If $A \sim W_p(n, \Sigma)$

If a is a fixed $p \times 1$ vector then,

$$a'Aa \sim (a'\Sigma a) X_n^2$$

Or equivalently, $\boxed{\frac{a'Aa}{a'\Sigma a} \sim X_n^2}$

↓

Wishart back to chi-square.

Proof: Start with, $A = \sum_i X_i X_i'$

Then, $a'Aa = a'\left(\sum_i X_i X_i'\right) a$

$$a'Aa = \sum_{i=1}^{n} a'X_i X_i' a$$

Since, $a'X_i X_i' a = (a'X_i)^2$

Therefore, $\boxed{a'Aa = \sum (a'X_i)^2}$

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Now, $X_i \sim N_p(0, \Sigma)$

Therefore the linear combination,

$a'X_i$ is Normally dist^n with $E(a'X_i) = 0$

$V(a'X_i) = a'\Sigma a$

&

$a'X_i \sim N(0, a'\Sigma a)$

Standardize $Z = \dfrac{a'X_i}{\sqrt{a'\Sigma a}} \sim N(0,1)$ $\longrightarrow$ $Z = \dfrac{X_i - \mu}{\sigma_{X_i}}$

Therefore $$\frac{a'Aa}{a'\Sigma a} = \sum_{i=1}^{n} \left(\frac{a'X_i}{\sqrt{a'\Sigma a}} \right)^2$$

But $\sum_i Z_i^2 \sim \chi_n^2 = \sum_i Z_i^2$

Thus, $$\boxed{\frac{a'Aa}{a'\Sigma a} \sim \chi_n^2}$$

$\mu = 14.$

$\bar{x}$.

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But $\sum_i Z_i^2 \sim \chi_n^2 = \sum_i Z_i^2$

Thus, $\boxed{\dfrac{a'Aa}{a'\Sigma a} \sim \chi_n^2}$

Sampling from a multivariate Normal dist^n & Maximum liklihood estimati:-

Suppose we take a random sample from a multivariate normal population. What informatio do we want to estimate?

$$X_1, X_2 \ldots X_n \overset{i.i.d}{\sim} N_p(\underset{\sim}{\mu}, \Sigma)$$

$\underset{\sim}{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_p \end{bmatrix}$ is the population mean vector

& $\Sigma \rightarrow$ Population cov. matrix.

Since μ & Σ are unknown, how can we estimate them from the sample?

This leads directly to MLE

Suppose $p = 2$, Each obs is, $X_i = \begin{bmatrix} X_{i1} \\ X_{i2} \end{bmatrix}$

Or our sample looks like

$$\begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \\ \vdots & \vdots \\ X_{n1} & X_{n2} \end{bmatrix}$$

Each col. is one variable

Each row is one individ

Maximum liklihood Idea:-

"We choose the parameter values that make the observed sample most likely."

For a single obsn —

$$f(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left[\frac{-1}{2}(x_i - \mu)' \Sigma^{-1} (x_i - \mu)\right.$$

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Since obs. are independent :-

$$L(M, \Sigma) = \prod_{i=1}^{n} f(x_i, M, \Sigma)$$

↓
likelihood function

Now, we need to find the values of M and Σ that maximize this likelihood.

$$l = \log(M, \Sigma) = \frac{-np}{2}\log(2\pi) - \frac{n}{2}\log|\Sigma| - \frac{1}{2}\sum_{i=1}^{n}(x_i - M)'\Sigma^{-1}(x_i - M)$$

(log L)

~~Now, diff this w.r.to M & Σ & set the derivation equal to zero~~

MLE of Mean vector :-

~~$$\frac{\partial l}{\partial M} = 0 - 0 - \frac{1}{2}\Sigma^{-1}\sum_{i=1}^{n}(x_i - M)^2$$~~

$$\frac{\partial}{\partial \underset{\sim}{M}}\log L \text{ or } \frac{\partial l}{\partial \underset{\sim}{M}} = -\frac{1}{2}\sum_{i=1}^{n}$$

Simplify l first,

$$l = \frac{-np}{2}\log(2\pi) - \frac{n}{2}\log|\Sigma| - \frac{1}{2}\sum_{i=1}^{n}\left[\underset{\sim}{x}_i\Sigma^{-1}x_i - 2\underset{\sim}{x}_i'\Sigma^{-1}\underset{\sim}{M} + \underset{\sim}{M}'\Sigma^{-1}M\right)$$

Now diff w.r. to M :-

$$\frac{\partial}{\partial \mu} l = \frac{-1}{2} \sum_{i=1}^{n} \left(-2 x_i' \Sigma^{-1} + 2 \mu' \Sigma^{-1}\right) = 0$$

$$= n \mu' \Sigma^{-1} = \left(\sum_{i=1}^{n} x_i'\right) \Sigma^{-1}$$

↓ constant

$$\Rightarrow \hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i = \bar{x}$$

$$\boxed{\hat{\mu} = \bar{x}}$$

$(1\times p)(p\times p)(p\times 1)$

$(1\times p)(p\times 1)$

1×1

Explanation: Take

$$\frac{-1}{2} \sum_{i=1}^{n} (x_i - \mu)' \Sigma^{-1} (x_i - \mu)$$

Consider $Q_i = (x_i - \mu)' \Sigma^{-1} (x_i - \mu)$

To make it easier - Write $B = \Sigma^{-1}$

Then $Q_i = (x_i - \mu)' B (x_i - \mu)$, we need $\frac{\partial Q_i}{\partial \mu}$

Now, Expand the quadratic form - it's a scalar (1×1)

$$(x_i - \mu)' B (x_i - \mu) = x_i' B x_i - x_i' B \mu - \mu' B x_i + \mu' B \mu$$

For a scalar $a' = a$

Because Σ is a covariance matrix -

$$\Sigma' = \Sigma \Rightarrow \boxed{B' = B}$$

$x_i' B \mu = (x_i' B \mu)'$

$x_i' B \mu = \mu' B' x_i$

$$(\Sigma^{-1})' = (\Sigma')^{-1} = \Sigma^{-1}$$

So the middle terms are equal.

$$x_i' B \mu = \mu' B x_i$$

$= \mu' B x_i$

Thus, $Q_i = x_i' B x_i - 2 \mu' B x_i + \mu' B \mu$

$\frac{\partial}{\partial M}(a'M) = a$, $\frac{\partial}{\partial M}(M'BM) = 2BM$ when B is symmetric

$a = BX_i$ × constants

Now, diff w.r to M —

$$\frac{\partial}{\partial M}(a'M) = a$$

$$\frac{\partial}{\partial M}(M'BM) = 2BM$$

when B is symmetric

$$\frac{\partial Q_i}{\partial M} = 0 - 2Bx_i + 2BM$$

$$= -2B(x_i - M)$$

$$= -2\Sigma^{-1}(x_i - M)$$

$x_i^2 \Sigma^{-1} = 0$

$$\frac{\partial Q_i}{\partial M} = -2\Sigma^{-1}(x_i - M)$$

$$\frac{\partial l}{\partial M} = \frac{-1}{2}\sum_{i=1}^{n} Q_i$$

$$= \frac{-1}{2}\left[\sum_{i=1}^{n}\{-2\Sigma^{-1}(x_i - M)\}\right]$$

$$\frac{\partial l}{\partial M} = \Sigma^{-1}\sum_{i=1}^{n}(x_i - M) = 0$$

Since Σ^{-1} is Non Singular

$$\Rightarrow \Sigma(x_i - M) = 0$$

$$\Rightarrow \Sigma x_i - nM = 0$$

$$\Rightarrow \hat{M} = \frac{1}{n}\Sigma x_i$$

$$\Rightarrow \boxed{\hat{M} = \bar{X}}$$

So, MLE of the multivariate mean vector is the sample mean vector.

MLE of Covariance Matrix :- ($\hat{\Sigma}$)

$X_1, X_2 \dots X_n \overset{i.i.d}{\sim} N_p(\underset{\sim}{\mu}, \Sigma)$

$\Rightarrow$ We already obtained $\boxed{\hat{\mu} = \bar{X}}$

$\hat{\Sigma} = ?$ $\rightarrow$ Now we keep μ fixed and estimate the cov. matrix Σ.

$$l(\mu, \Sigma) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log|\Sigma| - \frac{1}{2}\sum_{i=1}^{n}(x_i - \underset{\sim}{\mu})'\Sigma^{-1}(x_i - \underset{\sim}{\mu})$$

First simplify the Quadratic term :-

Define.

$\boxed{A = \sum_{i=1}^{n}(x_i - \mu)(x_i - \mu)'}$

x_i = constant
μ = const
Σ = Variable

$\downarrow$ when diff w.r.to Σ.

Now, notice that, $(x_i - \underset{\sim}{\mu})'\Sigma^{-1}(x_i - \mu)$ is a scalar.

look at the dimensions —

Suppose, we have p variables, Then,

$x_i - \underset{\sim}{\mu}$ is a $p \times 1$ col. vector.

$x_i - \underset{\sim}{\mu} = \begin{bmatrix} x_{i1} - \mu_1 \\ x_{i2} - \mu_2 \\ \vdots \\ x_{ip} - \mu_p \end{bmatrix}$, then its transpose $(x_i - \mu)'$ which is a $1 \times p$ row vector.

Also, Σ^{-1} is a $p \times p$ matrix

So, the multiplication has dim

$$\underbrace{(1 \times p)(p \times p)}(p \times 1)$$

$$(1 \times p)(p \times 1) = 1 \times 1$$

Therefore, $(x_i - \underset{\sim}{\mu})'\Sigma^{-1}(x_i - \mu)$ is a 1×1 matrix is simply a scalar.

We can write, a scalar as its trace —

$$x'Ax = tr(x'Ax)$$

$$tr(x'Ax) = tr(Axx')$$

↓ cycle property of Trace.

Let A be a $k \times k$ symmetric matrix & x be a $k \times 1$ vector then

$$x'Ax = tr(x'Ax) = tr(Axx')$$

Matrix Result

Therefore,

$$(x_i - M)' \Sigma^{-1} (x_i - M)$$

$$= tr\left[\Sigma^{-1}(x_i - M)(x_i - M)'\right]$$

Summing over i,

$$\sum_i (x_i - M)' \Sigma^{-1} (x_i - M) = tr(\Sigma^{-1} A).$$

So, the log likelihood becomes,

For this check last Page

$$l = \text{constant} - \frac{n}{2}\log|\Sigma| - \frac{1}{2} tr(\Sigma^{-1} A)$$

$$\frac{\partial l}{\partial \Sigma} = -\frac{n}{2}\Sigma^{-1} - \frac{1}{2}\left[-\Sigma^{-1} A \Sigma^{-1}\right]$$

Why? Left to Students

$$\frac{\partial l}{\partial \Sigma} = -\frac{n}{2}\Sigma^{-1} + \frac{1}{2}\Sigma^{-1} A \Sigma^{-1} = 0$$

$$= -n\Sigma^{-1} + \Sigma^{-1} A \Sigma^{-1} = 0$$

$$\Sigma^{-1} A \Sigma^{-1} = n\Sigma^{-1}$$

Multiply from the left by Σ:

$$A\Sigma^{-1} = nI$$

Then multiply from the right by Σ:

$$A = n\Sigma$$

Therefore, $\hat{\Sigma} = \frac{A}{n}$

$$A = \sum_i (x_i - \underline{\mu})(x_i - \underline{\mu})'$$

Since the MLE of μ is, $\boxed{\hat{\mu} = \bar{x}}$

Then $\boxed{\hat{\Sigma}_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})'}$

Or

$$\boxed{\hat{\Sigma}_{MLE} = \frac{(n-1)S}{n}}$$

$$S = \frac{1}{n-1}\Sigma(x_i - \bar{x})(x_i - \bar{x})'$$

$$= (n-1)S$$

$$\frac{\partial}{\partial \Sigma} \log|\Sigma| = \frac{1}{|\Sigma|} \cdot \frac{\partial |\Sigma|}{\partial \Sigma}$$

Jacobi's Formula

(Chain Rule)

then apply Jacobi's Formula –

It gives the derivative of a matrix determinant w.r.to the matrix –

$$\frac{\partial |\Sigma|}{\partial \Sigma} = \text{adj}(\Sigma)^T = \text{adj}(\Sigma) \quad (\Sigma \text{ is symmetric})$$

$$\Sigma^{-1} = \frac{\text{adj}(\Sigma)}{|\Sigma|} \rightarrow \text{By definition of inverse of a matrix}$$

$$\frac{\partial}{\partial \Sigma} \log|\Sigma| = \frac{1}{|\Sigma|} \cdot \text{adj}(\Sigma) = \Sigma^{-1}$$

(II) $\frac{\partial}{\partial \Sigma}\left[tr(\Sigma^{-1}A)\right] = -\Sigma^{-1}A\Sigma^{-1}$?

Proof: In Matrix Calculas -

$$d(\Sigma^{-1}) = -\Sigma^{-1}(d\Sigma)\Sigma^{-1}$$

$$d\left[tr(\Sigma^{-1}A)\right] = tr\left[d(\Sigma^{-1})A\right]$$

$$df = tr\left[-\Sigma^{-1}A\Sigma^{-1}d\Sigma\right] = tr\left(-\Sigma^{-1}(d\Sigma)\Sigma^{-1}A\right)$$

$$= tr\left[-\Sigma^{-1}A\Sigma^{-1}d\Sigma\right] \leftarrow \text{By cyclic Property of Trace}$$

Applying matrix diffn identity -

$$\left[\frac{\partial}{\partial \Sigma} tr(\Sigma^{-1}A)\right]^T = (-\Sigma^{-1}A\Sigma^{-1})^T$$

Assuming Σ and A are symmetric -

$$(\Sigma^{-1}A\Sigma^{-1})^T = -\Sigma^{-1}A\Sigma^{-1}$$

For a general function, $f(\Sigma)$ the diff is given by

$$df = tr\left[\left(\frac{\partial f}{\partial \Sigma}\right)^T d\Sigma\right]$$

$$f = \frac{\partial}{\partial \Sigma} tr(\Sigma^{-1}A) = (\Sigma^{-1}A\Sigma^{-1})$$